

Towards Integrated AI-Assisted Legal Reasoning: A Survey of Indian Legal NLP Across Five Capability Axes

Neeta Chavan, Mayur Rathod, Megh Yashwantkar, and Gajanan Barve

Department of Computer Science and Business Systems

JSPM's Rajarshi Shahu College of Engineering, Pune, Maharashtra, India

Abstract—Close to fifty million cases sit unresolved across India’s three-tier court hierarchy. That number alone makes a strong case for computational intervention, yet the legal AI tools available today are built in silos: retrieval engines that know nothing about prediction models, prediction models that ignore explainability, and explainability modules that never feed back into retrieval. We survey Indian legal NLP through five capability axes: precedent search, case fact structuring, argument graph construction, judicial outcome simulation, and what-if analysis. For each axis we catalogue representative methods, benchmark datasets, and evaluation practices. Our principal observation is blunt: no existing system ties all five axes together. Argument graph construction and counterfactual what-if reasoning are, for practical purposes, absent from the Indian legal AI landscape. We argue that progress on individual axes has reached a point of diminishing returns without a unifying pipeline that couples retrieval, structured reasoning, prediction, and interactive analysis.

I. INTRODUCTION

India’s courts are drowning. Roughly fifty million cases remain pending across district courts, high courts, and the Supreme Court, and the backlog grows faster than the judiciary can clear it. This is not a scheduling problem; it is a structural failure of access to justice that periodic administrative reform has proven unable to fix. Litigants wait years, sometimes a full decade, for resolution. Public trust erodes quietly, case by case. AI-assisted legal tools offer a way to augment, not replace, judicial reasoning at a scale that human capacity alone cannot match. Over one million Supreme Court and High Court judgments are now digitised [1], providing an empirical base that few other common-law jurisdictions can rival. We believe these conditions make India one of the most consequential, and still underexplored, testbeds for legal AI research.

The tools that do exist, however, talk past each other. A retrieval engine that surfaces relevant precedents knows nothing about the prediction model sitting one API call away. The prediction model, for its part, generates a disposition label without any awareness of how the retrieval module ranked its candidates. Explainability components, where they exist at all, bolt on after the fact. This fragmentation is not how judges reason. A judge does not first retrieve, then apply statute, then weigh facts; she does all three simultaneously, circling back when a new precedent reshapes the factual picture. Recent empirical evidence supports this intuition: coupling retrieval

with downstream reasoning outperforms strict pipeline decomposition [2], and case-based reasoning integrated into retrieval-augmented generation produces answers that are measurably more grounded in law [3]. The lesson is clear, or should be, that tight integration across capability boundaries is not optional.

We should be upfront about what this paper is. It is a survey, not a system. We make two contributions: a structured review of five capability areas in Indian legal NLP, and the explicit identification of the missing integrated multi-capability pipeline as the field’s central open problem. The five axes are: (1) precedent search, encompassing sparse and dense retrieval of prior judgments; (2) case fact structuring, including named entity recognition, rhetorical role labelling, and statutory section tagging; (3) argument graph construction, covering extraction of claims, evidence, and their relational structure; (4) judicial outcome simulation, predicting dispositions with natural-language explanations; and (5) what-if analysis, enabling counterfactual hypothesis testing over case facts. These axes, taken together, span the full reasoning chain from information retrieval to interactive deliberation.

The rest of this paper follows a straightforward path. Section II sets up background on the Indian legal ecosystem and the specific headaches it creates for NLP. Sections III through VII walk through each capability axis in turn. Section VIII pulls back to compare across axes, flagging integration gaps and recurring themes. Section IX takes up evaluation methodologies. Section X outlines open problems and future directions. Section XI wraps up.

II. BACKGROUND: THE INDIAN LEGAL ECOSYSTEM

India inherited its common law tradition from British colonial rule, and that inheritance carries real weight: judicial decisions are precedent, not merely advisory. Judicial reasoning here rests on three pillars that a court must hold in mind at once: the established facts, the applicable statutes (principally the Indian Penal Code and the Code of Civil Procedure), and binding prior rulings invoked through *stare decisis*. No competent judge consults these sequentially. A judgment forms at the intersection of all three: facts are read through the lens of statute, statutes are interpreted in light of prior rulings, and prior rulings are tested against the factual matrix of the case at hand. We contend that any computational framework that treats

TABLE I
FIVE CAPABILITY AXES IN INDIAN LEGAL AI: COVERAGE AND IDENTIFIED GAPS

Capability Axis	Description	Gap in Existing Literature
Precedent Search	Semantic retrieval of relevant prior judgments through sparse keyword-based methods, dense vector representations, and hybrid strategies applied to Indian case law corpora.	Retrieval systems function as standalone components; their outputs are not consumed by downstream prediction or reasoning modules, limiting their utility in end-to-end legal analysis.
Case Fact Structuring	Extraction and classification of structured elements from judgment text, including named entity recognition, rhetorical role labelling of sentence-level functional segments, and Indian Penal Code section tagging.	Structured representations produced by these models are not propagated to any subsequent reasoning stage, rendering them terminal outputs rather than intermediate features in a larger pipeline.
Argument Graph Construction	Identification and relational mapping of argumentative components, including claims, supporting evidence, cited statutes, and invoked precedents, into directed graph representations that capture the logical structure of judicial reasoning.	This capability is virtually absent in Indian legal AI research; existing work on legal argument mining is confined almost exclusively to Western jurisdictions such as the European Court of Human Rights.
Judicial Outcome Simulation	Prediction of case dispositions (acceptance or dismissal) accompanied by natural language explanations that articulate the reasoning underlying the predicted outcome.	Prediction models operate without access to retrieval-augmented context; generated explanations remain shallow and lack grounding in specific precedents or statutory provisions.
What-If Analysis	Counterfactual hypothesis testing that enables users to perturb specific case facts, statutory provisions, or precedent selections and observe the resulting shifts in predicted outcomes and explanatory rationales.	Entirely unexplored in Indian legal NLP; no existing dataset, model, or evaluation framework addresses counterfactual reasoning over Indian case law.

these pillars as independent inputs to a staged pipeline has already compromised its fidelity to how law actually works.

Processing Indian legal text introduces a set of headaches that researchers working on ECHR or U.S. federal case law rarely encounter. The most obvious is multilingualism. A single judgment may drift between English, Hindi, Urdu, and a regional language mid-paragraph, with no markup or translation cues, which is a nightmare for tokenisers trained on monolingual corpora. Formatting is another pain point. There is no standard template across Indian courts; judgments from different benches, different decades, and different levels of the hierarchy vary wildly in structure, citation conventions, and rhetorical organisation. Rule-based extraction breaks almost immediately. And then there is annotation scarcity. The LegalEval shared task laid this bare: even trained annotators disagreed substantially on rhetorical role labels for Indian legal text [4]. General-purpose encoders like base BERT are not up to the task either. InLegalBERT showed that domain-adapted pre-training on Indian corpora is essentially non-negotiable for competitive performance [5]. We suspect the community still underestimates how much domain shift matters in this setting.

The good news is that resource creation has accelerated sharply. ILDC gave the community its first large-scale benchmark: over thirty-five thousand Supreme Court cases annotated for judgment prediction and explanation [6]. PredEx raised the bar by providing expert-annotated explanations tied to specific textual spans within judgments [7]. NyayaAnumana, the most recent and largest corpus, spans multiple court levels and ships with its own domain-specialised language model [1]. These datasets have driven measurable gains on tasks like binary judgment prediction and rhetorical role classification. But here is the catch. Every one of them targets a single capability

axis in isolation. No existing work unifies retrieval, structuring, reasoning, prediction, and interactive analysis under one coherent architecture. That gap, the absence of integration, is what motivates the five-axis taxonomy we develop in the sections that follow.

III. PRECEDENT SEARCH

Everything downstream depends on retrieval. In common law systems, prior rulings are not merely informative; they are binding. Finding the right precedent is the first move a lawyer makes when building an argument, and it is the step that every other task in the pipeline inherits quality from (or is poisoned by). Grounding generation in retrieved case law produces markedly better outputs than letting a language model rely on its parametric memory alone [2], and pairing retrieval with case-based reasoning pushes legal fidelity further still [3]. We think it is fair to say that retrieval is the single highest-leverage point in any legal AI system: get it wrong, and nothing else matters.

A. *Keyword-Based and Sparse Retrieval*

For most Indian lawyers, precedent discovery still means typing terms into Indian Kanoon. The platform runs on term-matching, and the workhorse behind it is BM25, where documents are scored by term frequency scaled against inverse document frequency across the corpus. Simple as the formulation sounds, BM25 is stubbornly hard to beat. Empirical comparisons on legal retrieval benchmarks show it matching or outperforming several neural alternatives, especially when the query contains distinctive legal phrases [8]. We suspect that the community often reaches for neural approaches prematurely, without first exhausting what a well-tuned BM25 can deliver. The failure mode is precise but consequential:

vocabulary mismatch. A query about “breach of fiduciary duty” returns nothing when the relevant judgment phrases the same concept as “violation of trust obligations.” No amount of IDF weighting can bridge a gap where the surface terms simply do not overlap.

B. Dense Retrieval

Dense retrieval attacks the vocabulary mismatch problem head-on. Queries and documents are encoded into continuous vector representations, and similarity is computed via cosine or dot-product in the embedding space. Transformer-based encoders learn to push semantically related texts close together, so “breach of fiduciary duty” and “violation of trust obligations” end up in the same neighbourhood, even though they share no keywords. Dual-encoder architectures with summarisation-based re-ranking have proven competitive on the COLIEE case law retrieval task [9]. Scaling this to real corpora, however, is a different problem. FAISS provides the standard approximate nearest-neighbour infrastructure, enabling sub-linear search over billion-scale vector collections on commodity GPUs [10]. The cost is non-trivial: index construction and query-time inference both demand significantly more compute than BM25’s lightweight inverted indices. Whether this trade-off pays off for Indian legal corpora, which are large but not yet at the scale where sparse retrieval collapses, remains an open empirical question.

C. Domain-Adapted Models for Indian Legal Retrieval

A BERT checkpoint trained on Wikipedia and BooksCorpus knows very little about “ratio decidendi” or “Section 302 IPC.” Legal text is unusual enough (long sentences, nested citations, Latin-inflected phrasing) that general-purpose encoders consistently underperform on legal retrieval and classification tasks. The fix is straightforward in principle: continued pre-training on in-domain data. InLegalBERT does exactly this, pre-training a BERT-based model on a large corpus of Supreme Court and High Court judgments [5]. It consistently beats both vanilla BERT and multilingual variants across Indian legal benchmarks. We read this result as confirmation that domain shift in legal NLP is not a minor nuisance; it is a first-order problem that general-purpose scale alone does not solve.

D. Retrieval-Augmented Generation for Legal Reasoning

RAG has become the dominant pattern for grounding language model outputs in retrieved evidence, and legal reasoning is a natural fit. Instead of relying on whatever a large language model memorised during pre-training, the generative component receives actual precedents as context. NyayaRAG applies this directly to Indian judgment prediction and shows that retrieval-grounded generation substantially outperforms standalone generation across multiple court levels [1]. CBR-RAG goes a step further: rather than retrieving cases that are merely semantically close, it incorporates case-based reasoning to surface structurally analogous cases, improving the legal relevance of the context window [3]. These are meaningful advances. But we note a persistent structural limitation: in

every existing Indian legal AI system we have surveyed, retrieved outputs feed into generation modules and stop there. They are never routed to structured fact analysis, argument construction, or outcome prediction. Retrieval remains an endpoint rather than an input to a broader reasoning chain.

IV. CASE FACT STRUCTURING

An Indian Supreme Court judgment can run to tens of thousands of words. Procedural history, factual narrative, statutory analysis, counsels’ submissions, and the final order are all woven together without consistent formatting or explicit section markers. Feed that to a prediction model as raw text and you are asking it to find a needle in a haystack while blindfolded. We argue that structuring, that is, decomposing the document into its functional parts, is a prerequisite, not a convenience. Three structuring tasks have received sustained attention in Indian legal NLP: named entity recognition, rhetorical role classification, and statutory section tagging.

A. Legal Named Entity Recognition

General-purpose NER (person, location, organisation) is almost useless on a court judgment. The entities that matter are judge names, petitioner and respondent roles, case numbers, statute references, and cited precedents. Kalamkar et al. recognised this and built a fourteen-type schema designed specifically for Indian judicial text, covering courts, judges, petitioners, respondents, lawyers, dates, case numbers, statutes, provisions, precedents, witnesses, other persons, organisations, and geographical entities [11]. Transformer-based sequence labelling models were evaluated against this benchmark on Supreme Court and High Court judgments. The downstream payoff is immediate. Precedent entities can be linked to their source judgments for retrieval. Party entities feed coreference resolution. Statute entities bridge to the statutory knowledge base. Without this scaffolding, every downstream module is flying blind, forced to operate over raw token sequences with no structured hooks to latch onto. We find it surprising that legal NER still receives relatively little attention compared to classification tasks, given how much of the pipeline depends on it.

B. Rhetorical Role Classification

Every sentence in a judgment plays a role: stating facts, citing a prior ruling, framing a legal issue, articulating reasoning, or delivering the operative decision. Rhetorical role labelling assigns these functional categories automatically. LegalEval (SemEval-2023 Task 6) established a multi-court benchmark for this task across Indian jurisdictions, and the results were instructive: automated segmentation is feasible, but far from solved [4]. The hard part is context dependence. “The appeal is dismissed” means one thing when it appears inside a recitation of a prior ruling and something entirely different as the operative order of the present case. You cannot classify the sentence in isolation; you need long-range context. Hierarchical and sequential models that propagate information across sentence boundaries help, but we observe that most

current approaches still struggle at document lengths typical of Indian high court judgments. Rhetorical segmentation matters because it is the only way to isolate the ratio decidendi from the noise; without it, every sentence carries equal weight, which is obviously wrong.

C. IPC Section Tagging and Statute Linking

Statute identification asks a simple question: which sections of the Indian Penal Code (or other legislative instruments) does this judgment actually engage with? LegalEval includes it as one of three subtasks, with annotated data and evaluation protocols [4]. The task matters because it is the bridge between facts and law; without knowing which statutes apply, neither argument construction nor outcome prediction can proceed on solid footing. The persistent difficulty is implicit reference. Judges routinely discuss the elements, interpretation, and applicability of a statute without ever writing down the section number. The model must perform semantic matching between case language and statutory text, which is considerably harder than spotting an explicit “Section 302” mention.

Consider NER, rhetorical roles, and statute tagging together. They address complementary dimensions of document structure: who, what role, and under what law. All three should feed into downstream reasoning as intermediate representations. In practice, they do not. Each task has been studied in isolation within Indian legal NLP, and no existing system propagates their combined outputs into a unified pipeline. We are left with a structuring layer that produces terminal outputs, carefully extracted, then abandoned.

V. ARGUMENT GRAPH CONSTRUCTION

This is the axis where the literature thins out sharply. Legal reasoning is, at its core, argumentative. A decision crystallises from competing claims, supporting evidence, statutory authority, and precedent invoked by analogy or distinction. If you can represent these components and the relations between them (support, attack, citation, analogy) as a directed graph, you have made the inferential skeleton of a case explicit and machine-readable. We consider argument graph construction the most structurally ambitious of the five axes in this survey. It is also, by a wide margin, the one with the least Indian legal AI work behind it. What follows therefore draws heavily on the broader argument mining literature and on empirical work conducted in Western jurisdictions.

A. Argument Mining: Foundations

Argument mining extracts claims, premises, and evidence from text and maps the relations between them: support, attack, rebuttal. The most comprehensive legal application to date is Habernal et al.’s work on European Court of Human Rights decisions, where a taxonomy of seventeen argumentative component types was used to annotate and train transformer-based extractors [12]. Performance is reasonable on shorter decisions but degrades noticeably as documents grow longer and argumentative structure becomes more tangled. The field has moved from sentence-level classification (is

this sentence a claim or a premise?) to graph-level modelling, where nodes represent argumentative units and directed edges encode support, contradiction, and evidential relations. This shift matters enormously for legal applications. A single precedent might support a petitioner’s claim while simultaneously undermining the respondent’s, and only a graph can capture that kind of ambivalence.

B. Argument Structure in Legal Corpora

Almost all empirical work on legal argument mining has been conducted on European and American case law. LexGLUE, the closest thing to a standard benchmark for legal language understanding, evaluates primarily on ECHR and other Western corpora [13]. Indian courts are simply not represented. That is not a minor gap; it is a blind spot. The structure of a legal argument graph maps naturally onto the components of a judgment. Claims correspond to legal issues in dispute. Evidence nodes capture trial-established facts. Statute nodes provide the normative warrant. Precedent nodes supply analogical support or distinguishing authority. With a complete graph in hand, weak-point detection, defined as identifying claims that lack adequate evidentiary or statutory backing, becomes a matter of graph completeness analysis. We think this capability alone would be worth the annotation investment, even if no further downstream use were envisioned.

C. The Indian Legal Gap and Graph Database Representations

As of this writing, no published dataset, annotation schema, or trained model addresses argument graph construction for Indian legal judgments. The gap is total. Several factors explain it. Indian judgments are substantially longer and more digressive than the ECHR decisions that existing argument mining models were designed for. Their argumentative structure is less formulaic; reasoning is often scattered across non-contiguous sections, making span extraction unreliable. And the most fundamental barrier is the simplest: annotated training data for Indian courts does not exist.

On the representational side, graph databases like Neo4j are a natural fit. Nodes for claims, evidence, statutes, and precedents; typed edges for support, citation, contradiction, and analogy. The architecture almost designs itself. What makes this axis particularly compelling from a systems perspective is that its inputs already exist, in principle. Retrieved precedents from Section III, NER entities, rhetorical segments, and statute tags from Section IV all feed naturally into argument graph construction. This axis is the connective tissue of the pipeline. Without it, the outputs of retrieval and structuring have nowhere to go. We consider Indian legal argument mining a high-priority research direction, and we are frankly surprised it has received so little attention.

VI. JUDICIAL OUTCOME SIMULATION

Prediction has the longest research history in the field, moving from basic classifiers to the current LLM state-of-the-art. The task is straightforward: take the text of a case and

predict if it was accepted or dismissed. If you can, generate the reasoning too. We are currently at a transition point where the community is realizing that accuracy numbers, taken alone, are nearly meaningless in a legal context. If a model predicts the right outcome for the wrong legal reasons, it isn't a tool; it's a liability.

A. Classical and Early Neural Approaches

The early days were defined by manual feature engineering. Researchers spent months hand-crafting indicators for bag-of-words or TF-IDF models. They worked, sort of. But they were brittle, failing the moment they saw a case from a different court or a different decade. Neural networks, specifically CNNs and RNNs, removed the need for manual features but hit a different wall: document length. Indian judgments are long. Really long. Most early recurrent models simply ran out of memory or lost the context by the time they reached the operative order. LegalBERT was the first real breakthrough here, proving that domain shift is so severe in law that general-purpose embeddings are essentially a different language.

B. Transformer-Based Prediction

Transformers changed what was possible. By leveraging self-attention, models like InLegalBERT could finally capture the dependencies between a factual assertion on page two and a statutory application on page twenty [5]. The ILDC benchmark made this evaluation systematic for the first time, providing 35,000 cases to test these architectures [6]. However, the 512-token limit of base BERT remains a constant frustration. We have seen researchers try everything from overlapping chunks to sliding windows, but we suspect this ceiling is more damaging than current evaluations reveal. You cannot model a 100-page high court judgment by looking at it through a series of disconnected 500-word keyholes.

C. Explainability as a First-Class Requirement

A model that outputs "dismissed" without a reason is useless to a lawyer. You cannot audit it, you cannot identify its biases, and you certainly cannot build trust with it. PredEx addresses this by providing 15,000 cases where the rationales aren't just model-generated hallucinations, but are actual explanations written by legal experts [7]. This moves the goalpost from "does the model look right?" to "is the model grounded in expert reasoning?" We observe, however, that even these expert annotations are often compressed summaries. Does a summary truly capture the recursive, self-correcting nature of a judge's internal reasoning?

D. Large Language Models and the Scale Frontier

Scale has become the default answer to everything in NLP, and legal AI is no exception. NyayaAnumana and its specialized model, INLegalLlama, represent the current ceiling for Indian legal prediction [1]. These models internalize the rhythms of judicial language across 700,000 case proceedings. But scale has brought a new problem: benchmark bias. Many of our current test sets are cherry-picked from cases with

unambiguous outcomes, which makes the models look much better than they actually are in the messy reality of a contested courtroom [14]. We believe benchmark bias is the single most underappreciated methodological issue in the field today.

E. Remaining Limitations

Three gaps remain. First, prediction models are still built in isolation from the retrieval and structuring axes we discussed earlier. There is no pathway for a structured fact to inform a final prediction.

Second, the data is too Supreme Court heavy. District courts are where the real backlog lives, yet they have almost no presence in our benchmarks.

Third, the evaluation metrics are often lazy. Accuracy is the easiest thing to measure, but in law, it is also the most misleading. If your model achieves 90% accuracy but fails every time a witness is deemed unreliable, it isn't just imperfect; it is dangerous because nobody has checked those specific failure modes.

VII. WHAT-IF ANALYSIS

What-if analysis is the capability that practitioners actually want, yet there is almost no literature behind it. It is also, in our view, the one that matters most. While prediction tells you what the outcome is, and explainability tells you why, what-if analysis asks what would happen if the facts were different. For a lawyer preparing a case, counterfactual reasoning is the primary form of explainability that matters.

A. Counterfactual NLP

In standard NLP, this usually means perturbing an input to see when the label flips. If I change a "dowry demand" to a "loan request," does the conviction stand? This is not just a technical curiosity. It is a way to map the decision boundaries of a model and identify which facts are actually driving the outcome. We distinguish between forward-looking what-if analysis (testing hypotheses) and backward-looking counterfactuals (explaining past decisions).

B. XAI in Legal Decision-Making

The legal system has a transparency requirement that most AI systems cannot meet. You have a right to know why your interest was affected, especially if an algorithm helped decide it [15]. Most current XAI methods like LIME or SHAP are designed for feature importance, not the structured counterfactuals that legal reasoning requires. Until systems can answer "what if," their explanations will remain descriptive rather than prescriptive.

C. The Indian Gap

There is no what-if analysis system for Indian law. We have found zero published papers on the topic. This is a massive opportunity for the community, especially given the resource constraints in Indian litigation.

A tool that lets a lawyer test different statutory framings or fact patterns without needing a new dataset every time would

be transformative. It is the logical endpoint of the integration we are proposing.

Our survey ends here because the literature ends here. We cannot review what hasn’t been built, but we can point to the empty space and call it a priority.

VIII. DATASETS

Datasets have been the bottleneck. Western legal AI had the luxury of inheriting decades of structured case law from repositories like LexisNexis and PACER; Indian legal NLP has had to build its benchmarks from scratch, one annotation campaign at a time, typically relying on legal professionals whose time is expensive and scarce. We survey the seven principal datasets available as of 2025. They differ sharply in scale, court coverage, annotation depth, and public availability, and understanding those differences is essential for evaluating where the field actually stands versus where it claims to be.

The jump from ILDC’s 35,000 cases to NyayaAnumana’s 700,000 is not incremental; it is a qualitative shift that makes LLM-scale fine-tuning feasible for Indian legal text for the first time [1]. But scale and depth are not the same thing. NyayaAnumana gives you outcome labels and court-level metadata; it does not give you the expert-written natural language explanations that PredEx provides at a fraction of the size [7]. We would argue that any claim about “state-of-the-art” Indian legal judgment prediction needs to specify whether the model was trained with explanation supervision or without it, because the two settings produce fundamentally different systems. District court data, meanwhile, is almost invisible across all annotated datasets, a glaring omission given that district courts handle the vast majority of India’s pending caseload. The InLegalBERT corpus and Indian Kanoon offer raw text at scale, but with no task-specific annotations their use is limited to pre-training and unsupervised exploration.

Three gaps stand out, and none of the seven datasets we surveyed addresses any of them. There are no argument-level annotations linking claims to evidence, statutes, and precedents within a single judgment. Without them, argument graph construction (Section V) is forced to operate in an unsupervised or weakly supervised mode. There are no multilingual annotations that handle the Hindi–English or regional-language code-mixing that pervades Indian judicial text, despite the fact that this code-mixing is not rare; it is routine. And there are no counterfactual annotations of the kind a what-if analysis system (Section VII) would need for training or evaluation. Each of these gaps maps directly onto an open research direction we have flagged earlier. Closing them will require annotation campaigns that bring NLP researchers and legal domain experts together in ways the community has not yet managed at scale [4].

IX. EVALUATION METRICS

How you measure a system matters at least as much as how you build it, and this is a point that legal AI papers tend to gloss over. Pick the wrong metric and you end up optimising for something the legal system does not care

about, or worse, producing a model that looks excellent on a leaderboard but fails catastrophically on the cases that matter most. The majority of legal AI studies report accuracy as their sole measure, without justifying that choice or acknowledging what it hides [16]. On imbalanced datasets (and Indian legal datasets are almost always imbalanced) accuracy is close to meaningless. We survey the metrics used across the retrieval, classification, and generation tasks covered in this paper, and where appropriate, we flag where metric choice has led the field astray.

A. Retrieval Metrics

MAP (Mean Average Precision) remains the default for ranked retrieval evaluation. It computes precision at each relevant document in the ranked list, averages across the list, and then averages again across queries, yielding a single number that captures both how many relevant documents were retrieved and how high they were ranked. nDCG (Normalised Discounted Cumulative Gain) complements MAP by applying a logarithmic discount to lower-ranked results, which aligns well with how lawyers actually search: they look at the first page, maybe the second, and rarely go further. Both BM25 evaluations [8] and dense retrieval experiments in COLIEE [9] report MAP and nDCG as primary measures.

The catch, and we think this is underappreciated, is that both metrics require relevance judgments. In legal retrieval, relevance is not a surface-level property. Whether a 1987 Supreme Court ruling on tenant eviction is “relevant” to a 2022 high court dispute over landlord obligations is a question of legal interpretation, not keyword overlap. Collecting those judgments at scale requires legal expertise, which makes evaluation itself bottlenecked by the same annotation scarcity that constrains training data.

B. Classification, Generation, and Explainability Metrics

Accuracy and macro-averaged F1 are the standard pair for binary and multi-class outcome prediction. On balanced data, accuracy is fine. Indian legal datasets are rarely balanced. Dismissals dominate most corpora, which means a classifier that predicts “dismissed” every time can post a high accuracy score without having learned anything at all [16]. Macro-F1 corrects for this by weighting each class equally, and we would argue it should be the primary metric in every legal judgment prediction paper, not a secondary one buried in an appendix.

NER evaluation requires a finer granularity. Aggregate F1 across all entity types can hide the fact that a model performs well on common categories (dates, court names) while completely failing on rare but legally critical ones like witness names and provision numbers [11]. Per-entity-type F1 is the minimum standard; anything less is misleading.

Explanation evaluation is harder, because the thing being measured, legal soundness, is not something an automated metric can assess. ROUGE and BERTScore capture surface overlap with reference text, which tells you whether the generated explanation uses similar words. It tells you nothing

TABLE II
PRIMARY DATASETS FOR INDIAN LEGAL NLP RESEARCH (2021–2025)

Dataset	Reference	Size	Court Coverage	Annotation Type	Avail.
ILDC	[6]	35,000 cases	Supreme Court of India	Binary outcome labels, multi-sentence explanations	Public
PredEx	[7]	15,000+ cases	Supreme Court of India	Expert-annotated natural language explanations aligned to outcomes	Public
NyayaAnumana	[1]	700,000+ cases	Supreme Court, High Courts, District Courts	Outcome labels, court-level metadata	Public
NyayaRAG Corpus	[2]	Subset of Indian judgments	Supreme Court and High Courts	Retrieval-annotated with relevant precedent links	Partial
LegalEval (SemEval-2023 Task 6)	[4]	9,380 sent. (NER); 14,864 sent. (roles)	Supreme Court and High Courts	NER labels (14 types), rhetorical role labels, statute tags	Public
InLegalBERT Corpus	[5]	5.4M sentences	Supreme Court and High Courts	Pre-training corpus, no task-specific labels	Partial
Indian Kanoon	–	Millions of documents	All court levels	No annotation, raw text only	Public

about whether the reasoning is correct. Expert Likert-scale assessments, where legal professionals rate adequacy, accuracy, and completeness on structured scales, remain the only reliable approach. Both INLegalLlama [1] and PredEx [7] adopted this methodology. It is expensive. It does not scale well. And it is, as far as we can tell, irreplaceable, which is a problem the community has not yet confronted honestly.

X. GAPS AND CONTRIBUTION

We can state the central finding plainly: nobody has built it yet. No existing Indian legal AI system connects precedent retrieval to case fact structuring, structuring to argument graph construction, argument graphs to outcome prediction, or prediction to counterfactual what-if analysis. Each axis we surveyed operates in its own silo. Retrieved precedents are never routed to a structuring module. Structured facts and extracted entities stop at the output layer; no argument graph constructor picks them up. Prediction models run without argument-level context, and their outputs are never fed into any counterfactual reasoning component. The closest the field has come to integration is coupling retrieval with generation [2] or with case-based reasoning [3], but even those systems touch only two of the five axes. We think this fragmentation is no longer a reflection of the field’s youth; it is a design choice the community has made by default, and one that needs to be made deliberately instead.

At the task level, three gaps stand out. Argument graph construction for Indian legal judgments has no published dataset, no trained model, and no evaluation benchmark. It is, by any measure, the largest structural blind spot in the field. Counterfactual what-if analysis has not been attempted at all in the Indian context, even though it is arguably the capability most relevant to practising lawyers, the people these systems are ostensibly being built for. And evaluation remains systematically tilted toward Supreme Court data [6],

[7]. District and high courts, which together account for the overwhelming bulk of India’s pending caseload, lack reliable benchmarks. LegalEval’s annotation scope reflects this same imbalance [4].

The data-level picture is no better. No annotated resource captures argument-level structure, specifically the claim-evidence-statute-precedent linkages that argument graph construction would require. Multilingual annotations are absent entirely; Hindi and regional language passages embedded in English judgments remain invisible to every model we surveyed. The large-scale corpora that enable LLM fine-tuning, NyayaAnumana chief among them [1], trade annotation depth for volume. PredEx offers expert-level explanations but at a fraction of the scale [7]. No single resource delivers both. We suspect that until one does, the field will continue producing systems that are either large-scale but shallow, or deep but narrow.

We should be clear about what this paper contributes and what it does not. It is a survey. We provide the first structured review of Indian legal NLP organised around five functional capability axes, map the existing literature to each axis, catalogue the datasets and evaluation methods in use, and identify what remains unsolved. Our primary claim is that individual capabilities have matured enough to make integration feasible, and that the absence of integration is now the binding constraint on meaningful progress. We hope this survey serves as a useful reference for anyone aiming to build toward a unified Indian legal AI architecture, but we make no claim that building one will be easy.

XI. ETHICAL CONSIDERATIONS

Train a model on historical Indian court judgments and it will learn what those judgments contain, including their biases. Outcome disparities correlated with gender, caste, economic status, and geography are well documented in Indian legal

history. A prediction system optimised to reproduce historical outcome distributions will reproduce historical inequities along exactly these lines, unless someone intervenes during training and evaluation. The problem is not unique to India; bias amplification has been flagged across jurisdictions [16]. But the stakes escalate sharply when model outputs start influencing real legal decisions, because there is a genuine risk that statistical artefacts get mistaken for legal reasoning by users who trust the system more than they should. We believe bias auditing across demographic and geographic subgroups should be a non-negotiable prerequisite for deployment, not a post-launch afterthought.

Explainability, in a legal setting, is not a feature. It is a procedural obligation. Parties to litigation have a right to understand the basis of decisions that affect their legal interests, and that right does not evaporate when part of the decision process is automated [15]. This creates a real problem for the field, because the models that predict best are the ones that explain least. A generated paragraph that reads fluently and cites the right statutes may still be legally incoherent, and no automated metric will catch that. PredEx’s approach, grounding explanation evaluation in expert annotation rather than ROUGE scores [7], is the right direction. But the community needs to internalise a harder lesson: fluency is not soundness. A model that writes well but reasons poorly is more dangerous, not less, because its confidence is convincing.

One distinction cannot be blurred. Everything we have surveyed in this paper, including retrieval systems, structuring modules, prediction models, and explanation generators, belongs in the category of decision-support tools. They organise information, surface precedents, structure arguments, and estimate outcomes so that human legal professionals can reason more effectively. They are not autonomous adjudicators, and positioning them as such would be irresponsible. The most capable systems currently available still exhibit significant limitations in coverage, interpretability, and generalisation that rule out unsupervised deployment [1], [17]. Building Indian legal AI responsibly will require sustained collaboration among NLP researchers, legal professionals, and policymakers. The goal is augmentation, not automation, and losing sight of that distinction is the fastest way to erode the procedural safeguards that the rule of law depends on.

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